

Institutional Research Platform — Plan 3 of 3: Execution

Companion to Plan 1 (Foundations) and Plan 2 (Methodology) Date: 2026-08-16 · Status: Phase 0 deliverable — awaiting owner approval **Budget:** 2–3 h/day (owner decision Q5) · **No deadline** (Q4)

1. Answer to owner question Q2 — is the signal ledger required?

1.4 The four registered studies (owner decision, 2026-08-16)

2. Phase plan

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4. What I have versus what I need

5. Open questions

6. Risks

7. Definition of done for v1

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1. Answer to owner question Q2 — is the signal ledger required?

The owner asked me to check whether `institutional_signal_ledger` and `experiment_registry` are needed given the research-only scope, and to answer based on what produces a good-quality outcome.

They are not the same kind of object, and the answer differs.

1.1 `experiment_registry` — REQUIRED, build it first

It records a hypothesis, its pass bar and its kill criteria **before** the data is looked at. Without it, every number the platform produces is exploratory, and an exploratory number cannot be distinguished from a number that was tuned until it looked good.

This matters more here than in most projects for a specific reason: **an exploratory pass has already been run.** The audit measured the institutional premise on 2026-08-16 and found $t \approx -0.8$ at 1 and 3 months. Any subsequent study is now, strictly speaking, a second look at data whose answer is partly known. The only way to keep the result credible is to register the full specification — horizons, cost policy, benchmark, correction method, pass bar — before re-running, and to record in `exploratory_prior_run` that the earlier look happened and what it showed.

Without the registry, the honest description of this project's output would be "we looked at the data many ways and reported what we found." With it, the description is "we declared what would count as evidence, then measured it." That distinction is the quality of the outcome.

Verdict: build it in Phase 1, before any research code.

1.2 `institutional_signal_ledger` — NOT required now

Its purpose (original plan §14) is to record **engine decisions**: which engine approved or rejected which signal, on what date, for what reason. With no engines, it would be a permanently empty table.

What research actually needs in its place is a **result store** — `study_result` (Plan 1 §7.4) — recording each stratum's N , effect, corrected p , correction method, family size, bootstrap CI, input hashes and verdict. That is a different shape entirely: one row per *finding*, not per *decision*.

Verdict: create the table empty in migration 0001 so engines drop in later without a schema change (owner decision Q1), but do not populate or maintain it in v1. `study_result` is what carries the research.

1.3 What this means practically

Table	Phase	Populated in v1?
<code>experiment_registry</code>	1	✓ Yes — every study
<code>study_result</code>	6	✓ Yes — every finding
<code>artefact</code> / <code>artefact_edge</code> (provenance DAG)	1	✓ Yes — everything
<code>engine_config</code>	1	■ Created, all rows DISABLED
<code>institutional_signal_ledger</code>	1	■ Created, stays empty (test asserts it)

1.4 The four registered studies (owner decision, 2026-08-16)

All four are locked and run in Phase 6, in descending order of expected value. Each is a separate experiment with its own spec hash; registering four means the multiple-testing correction accounts for four families, raising each bar slightly. That is the accepted price for four honest answers.

#	Study	Question	N	Prior look?
1	Consensus	Do 3+ unrelated institutions buying the same name within 21 sessions predict outperformance?	pooled	No
2	Selling	Do disclosed institutional sales predict underperformance?	34,270	No
3	Block deals	Do negotiated blocks (0.7% round-trip) behave differently from bulk?	12,430	No
4	Bulk buys	The original premise, run properly	30,771	Yes — must disclose

Why this order. Consensus is statistically strongest: single-participant skill is hopeless (SBI Mutual Fund has 80 buys in 20 years — skill and luck are indistinguishable at that N), but a pooled convergence event needs no individual institution to be smart. Selling is 34,270 completely unexamined events with a plausible mechanism — institutions buy for many reasons (inflows, index tracking, rebalancing) but sell for fewer. Blocks are the cleanest data but only ~620 events/year. Bulk buys already look dead at $t \approx -0.8$; study 4 exists to close it honestly rather than leave it informally dismissed.

Study 4 carries a mandatory disclosure. Its `exploratory_prior_run` field records that a pass was run on 2026-08-16 and exactly what it found. A registration that conceals a prior look is not a registration.

2. Phase plan

Nine phases. Each ends with a deliverable and a gate; a gate that fails stops the next phase rather than being noted and passed.

Phase 0 — Audit and specification · COMPLETE

These three documents. Deliverable: integration map, schema proposal, feasibility findings, methodology, execution plan. **Gate: owner approval.**

Phase 1 — Skeleton, migrations, warehouse rebuild · ~3 weeks

Step	Detail
1.1	Freeze MICCV2 — unload 3 launchd agents, move plists, tag <code>frozen-2026-08-16</code> (Plan 1 §3.3)
1.2	Repo scaffold, <code>pyproject.toml</code> , Python 3.14 via uv, pytest, GitHub Actions running tests only (Q53)
1.3	<code>src/common/</code> : resolved repo root, config loader, structured logging, explicit-env guard, SHA-256 helpers
1.4	Trading calendar ported from MICCV2 — observed sessions, never generated. Gate: 5,339 sessions reproduced exactly
1.5	Migration runner: forward-only SQL files, checksummed, <code>schema_migrations</code> table
1.6	<code>0001_init.sql</code> — every table in Plan 1 §5–§7 and Plan 2 §2, §7.3, §8.1
1.7	Copy <code>v1_export</code> (1.2 GB) into <code>data/raw/v1_export/</code> , hash every file into <code>artefact</code>
1.8	Rebuild the price spine, adjusted spine, and PIT universe from <code>v1_export</code> with new code
1.9	Provenance DAG live from this point — every table written registers its artefact and edges
1.10	Close Risk 8 — <code>restic</code> to the 256 GB pendrive + free-tier cloud, plus a restore drill that is actually watched succeeding (§6)

Gate: the Plan 1 §3.4 reconciliation passes exactly — 7,749,148 price rows · 4,200 symbols · 1,497 dead symbols · 174,616,363 F&O rows · 5,339 sessions. Any mismatch is investigated before Phase 2.

Phase 2 — Raw archive and collection · ~3 weeks

Step	Detail
2.1	<code>src/archive/</code> : fetch → SHA-256 → dedupe on hash → store gzipped original → parse → parquet → record (Q11)
2.2	Browser-like session, 1 req/2 s, backoff, honest User-Agent (Q10)
2.3	NSE bulk + block parsers, historical archive endpoint (Q7)
2.4	Backfill 2026-07-09 → present (Q8) — the 5-week gap
2.5	Backfill the full NSE history through the archive endpoint, hashing every file
2.6	BSE bulk + block — new, never collected (Q9)
2.7	FII/DII cash collector — starts accruing forward (Q4 route: both)
2.8	<code>participant_oi</code> ported as the FII/DII proxy , labelled as F&O positioning, not cash flow
2.9	launchd agents installed and running from day one (Q13)
2.10	Measure available_from empirically — record observed publication time daily
2.11	Revision detection (Plan 1 §5.4)

Gate: 7 consecutive days collecting cleanly, every file hashed, zero overwrites, publication times recorded.

Phase 3 — Identity layer · ~4 weeks · *the highest-value phase*

Step	Detail
3.1	<code>security_master</code> wrapping the existing ISIN masters (Q15)
3.2	<code>symbol_history</code> — one <code>resolve(symbol, exchange, on_date)</code> function, all callers use it
3.3	Delisting detection from last-trade dates; classify MERGER / ACQUISITION / SUSPENSION
3.4	Resolve the 7,354 unmatched deal symbols — the single biggest quality win
3.5	<code>sector_history</code> PIT sectors from <code>index_membership</code> (13,163 rows with validity dates) (Q31)
3.6	Participant normalisation — aggressive cleaning, exact match only, no fuzzy auto-merge (Q19)
3.7	Behavioural classifier first: ≥95% same-day round-trip over ≥20 client-stock-days → <code>PROP_HFT</code>
3.8	Name-pattern classifier for the residual
3.9	Merge <i>suggestions</i> recorded, never applied; owner decides later (Q20)
3.10	Review queue CLI for the 1,515 names needing manual review
3.11	SHP collector → <code>promoter_entities</code> ; PIT promoter flag (Q25)
3.12	Manual fund-house mapping file (Q21)

Gate: unresolved symbol rate < 5% on the deal set — measured, not asserted. This gate is what turns the exploratory study into a defensible one.

Phase 4 — Clean mart · ~2 weeks

Step	Detail
4.1	<code>institutional_deals_clean</code> with all flags (Plan 1 §7.1)
4.2	Duplicate grouping — NSE/BSE cross-listing, both kept (Q24)
4.3	Same-day and 5-day round-trip flags (Q23)
4.4	Internal-transfer and promoter-related flags
4.5	Size eligibility: <code>deal_value_to_adv20</code> $\geq 0.5\%$ and $\geq ₹1\text{cr}$, configurable (Q26)
4.6	<code>eligible_for_research</code> excluding <code>PROP_HFT</code> by default, flag-driven (Q27)
4.7	Three interpretations — individual / accumulated / confirmation (plan §10)

Gate: every clean deal either resolves to a security or carries an explicit failure status. Zero silent drops.

Phase 5 — Cost model and benchmarks · ~2 weeks

Step	Detail
5.1	fee_schedule versioned table, rebuilt not ported — MICCV2's fee layer was wrong in 3 places (Plan 2 §4.1). Rates sourced from published rate cards and exchange circulars, every row carrying <code>verified</code> and a <code>source_url</code>
5.2	Corwin–Schultz and Abdi–Ranaldo spread estimators, rolling 21 sessions
5.3	Square-root market impact, <code>Y</code> configurable, sensitivity at 0.5 / 0.8 / 1.0
5.4	Participation cap and delay cost
5.5	Volatility-regime multiplier from India VIX
5.6	Six benchmarks incl. constructed smallcap and characteristic-matched (Plan 2 §5)
5.7	Gross / base / pessimistic reporting at every level

Gate: the cost model reproduces MICCV2's turn-of-month gross edge (+129.70 bps/occurrence) **and then correctly kills it**. MICCV2 scored it at 126 bps cost → +3.70 bps net → "survives". With the fee errors fixed the cost is 136.06 bps → **−6.36 bps net** → **dies**.

The gate asserts the *corrected* result. This is deliberate: a regression test that reproduces a wrong answer pins the bug. Reproducing the gross number proves the pipeline is faithful; flipping the net sign proves the fee fix is live.

Phase 6 — The outcome study · ~3 weeks

Step	Detail
6.1	Register all four experiments first (§2.1 below), each with its own spec hash, incrementing the trial counter to 72
6.2	Power analysis per stratum, before any fit (Plan 2 §6.5)
6.3	<code>deal_forward_outcomes</code> across 9 horizons × 6 benchmarks
6.4	Delisting/merger handling at 3 recovery factors (Q32)
6.5	Monthly-cohort collapse, moving-block bootstrap, NW-HAC
6.6	Three-scheme walk-forward: anchored (6) + rolling (11) + CPCV (66 paths)
6.7	PBO from the CPCV distribution
6.8	Romano–Wolf stepdown for participant/stratum ranking
6.9	Null-calibration: identical procedure on shuffled participant labels
6.10	Write <code>study_result</code> with corrected p , family size, CI, input hashes

Gate: every `study_result` row has a non-null `correction_method` and `n_tests_in_family`. Enforced by the schema, verified by a test.

Phase 7 — Seasonality rebuild and validation · ~3 weeks

Step	Detail
7.1	Atlas rebuilt from scratch, 13 windows × 4 alignments × 2 bases (Q42)
7.2	Observation minimums ≥ 10 yearly / ≥ 30 monthly (Q41)
7.3	Index expansion 46 → ~202 with dedup and history eligibility (Q40)
7.4	Near-duplicate grouping incl. return-correlation > 0.9 (Q36)
7.5	BY + BH + Storey q , over the actual run test count (Q37)
7.6	Permutation, 1,000 rotations (Q38)
7.7	Hansen SPA for best-of-family
7.8	Three-scheme OOS + full cost model

Gate: the run records its own `n_tests_in_run`. No hard-coded 31,893,556 anywhere — a test greps for it.

Phase 8 — Monitoring and reports · ~2 weeks

Daily / weekly / monthly markdown reports (Q51), DQ gates, pause logic, and the verification suite built to the Plan 2 §9.2 rules. Dashboard deferred.

Phase 9 — Deferred

Engines, paper portfolios, ML, LLM assistant. Not in v1.

3. Schedule

At 2–3 h/day (Q5), averaging $2.5 \text{ h} \times 6 \text{ days} \approx 15 \text{ h/week}$.

Phase	Weeks	Cumulative	Calendar (from 2026-08-17)
1 — Skeleton + warehouse	3	3	→ 2026-09-06
2 — Archive + collection	3	6	→ 2026-09-27
3 — Identity	4	10	→ 2026-10-25
4 — Clean mart	2	12	→ 2026-11-08
5 — Costs + benchmarks	2	14	→ 2026-11-22
6 — Outcome study	3	17	→ 2026-12-13
7 — Seasonality	3	20	→ 2027-01-03
8 — Monitoring + reports	2	22	→ 2027-01-17

~5.5 months to a complete research platform with its first answer. The first *publishable* result — the institutional outcome study — lands at the end of Phase 6, around mid-December 2026.

No deadline was set (Q4), so this is an estimate, not a commitment. The gates matter more than the dates: a failed gate stops the phase.

4. What I have versus what I need

4.1 Have — no action required

21-year price history including 1,497 dead symbols · 174.6M F&O rows · 223,450 bulk deals · 12,430 block deals · verified trading calendar · Nifty 500 TR benchmark · `participant_oi` FII/DII proxy 2014-2026 · index membership with validity dates · itemised statutory cost constants · working PDF toolchain · Python 3.14 + uv environment.

4.2 Need to build — no blockers

Raw archive · NSE and BSE collectors · identity layer · PIT sectors · promoter list · clean mart · cost model · benchmarks · outcome study · seasonality rebuild · monitoring.

4.3 Need from outside — real gaps

Gap	Impact	Route
Smallcap index history — 15 rows exist	One of six benchmarks	Construct from the price spine (Plan 2 §5.2), documented as constructed
FII/DII cash history — 22 days	Engine E deferred anyway	Unobtainable retrospectively; accrues forward from Phase 2
available_from for historical deals	Timing precision pre-2026	Measured going forward; conservative bound + LOW confidence for history
Parent fund-house relationships	Engine B deferred anyway	Manual mapping file (Q21)
Owner time for 1,515 name reviews	Blocks Phase 3 gate	~3–4 h in the review CLI, batchable

4.4 Need from the owner — decisions, not work

The open questions in §5, plus the Phase 3 review session.

5. Open questions

5.1 Q33 re-asked — excursions

The original question was unclear, so here it is plainly.

When an event is held for 12 months and ends **+20%**, that single number hides the path. Two very different histories produce it:

Path A:	+2% ... +8% ... +14% ... +20%	never below entry
Path B:	−41% ... −22% ... +5% ... +20%	down 41% at the worst point

Maximum Adverse Excursion (MAE) is the worst point (−41% in Path B). **Maximum Favourable Excursion (MFE)** is the best point reached along the way.

They matter because a strategy nobody could actually hold is not a strategy. If institutional-follow events routinely draw down 40% before recovering, the +7.80% headline is unreachable in practice.

The question: measure the worst point using *closing* prices only, or using *intraday* lows?

- **Closing prices** — what a daily-monitoring holder would have seen. Milder.
- **Intraday low** — the true worst tick. More severe, and it is what a stop-loss would have hit. Your OHLC data supports it.

My recommendation: intraday, because it is the honest worst case and understating drawdown is the more dangerous error. Reported alongside the close-based figure so both are visible.

5.1b Resolved by the 2026-08-16 external review

Item	Resolution
STT on delivery	0.1% both sides , not sell-only. MICCV2 under-charged 10 bps/round-trip. Turn-of-month's net edge flips negative
Exchange transaction charge	NSE 0.00307% , BSE 0.00375% — exchange-specific, and time-varying
GST base	18% of (brokerage + SEBI + transaction) , not brokerage alone
Fee constants	Replaced by a versioned fee_schedule with <code>source_url</code> and <code>verified</code> per row
Book-to-market for DGTW	Unavailable pre-2022 . Char-match becomes size × momentum × volatility × industry; BTM is a 2022+ sensitivity
CPCV group count	Fixed N=12 was wrong — purging a 24-month label exceeds a 20.6-month group. N now varies by horizon (190/120/45/15 paths)
PBO	CSCV logit-λ formulation specified explicitly
Hansen SPA	Studentised statistic + sample-dependent null; White's RC retained alongside for comparability with MICCV2
CS/AR estimators	Log prices, zero-range exclusion, negative-to-zero, 15% winsorisation, overnight adjustment; both estimates + ratio stored
ADV / vol windows	Explicit config with alternates reported
Participation cap	10% base and 5% conservative , the latter primary for the smallest tier
PROP_HFT thresholds	95% / 20 justified from the observed bimodal distribution; 9-combination sensitivity table published
Review queue order	By <code>deal_value × contribution to unresolved rate</code> , so the Phase 3 gate can pass before the queue empties
Risk 8	Closed in Phase 1.10 — restic to pendrive + free-tier cloud, with a watched restore drill

5.2 New questions arising from the audit

Resolved 2026-08-16: delisting recovery — all three, headline 0.0 · excursions — intraday primary, close alongside · studies — all four registered (§1.4) · brokerage — 0.03% headline with 0% reported alongside.

Still open — the defaults below apply unless the owner says otherwise:

- Consensus window.** What counts as consensus — 3+ institutions within 5, 21, or 63 sessions?
Default: all three as pre-declared variants inside study 1, with the correction accounting for three.
- Consensus threshold.** 3 institutions, or 2, or 5? *Default: 3, with 2 and 5 as declared variants.*
- Accumulation gap.** What gap closes an accumulation sequence? *Default: 63 sessions, configurable.*
- Repo public from day one (Q54), or at Phase 4?** Public now means negative findings are visible as they emerge — honest, but exposes half-finished work. *Default: public from Phase 1, README stating plainly it is in progress.*
- MICCV2 on disk after the Phase 1 gate?** Once the warehouse reconciles, its remaining value is its git history. *Default: keep until Phase 6, then tarball.*

6. Risks

#	Risk	Likelihood	Impact	Mitigation
1	The answer is no — no institutional edge exists	High	Low for the platform, high for expectations	Stated up front (Plan 2 §10). The platform's value is the ability to prove it
2	Symbol resolution stays poor, study stays biased	Medium	High	Phase 3 gate blocks at >5% unresolved
3	NSE/BSE change format or block collection	Medium	Medium	Raw bytes archived before parsing; a parse failure never loses the day
4	<code>available_from</code> unprovable for history	High	Medium	Conservative bound + LOW confidence flag; forward measurement from Phase 2
5	1,515 manual reviews stall	Medium	Medium	78% is already automatic; review is batchable and the queue is prioritised by deal volume
6	Scope creep back toward engines	Medium	High	Engines are schema-only, <code>DISABLED</code> , with a test asserting the ledger stays empty
7	Rebuilding the 31.9M atlas is slow	Medium	Low	Chunked by entity, resumable, checkpointed
8	Everything on one disk, no off-machine backup	High	Severe	Carried over from MICCV2's risk register unresolved — see below

Risk 8 deserves its own line, and it now has a concrete plan. MICCV2 recorded "everything is on one disk, no off-machine backup" as the highest-likelihood severe risk in its register and never closed it. The new repo inherits that exposure the moment `v1_export` is copied: 1.2 GB of irreplaceable history on a single laptop.

Closed in Phase 1, step 1.10, using the owner's 256 GB pendrive plus a free cloud tier. Two independent failure domains, ~2 GB to protect.

Tier	Target	Cadence	Contents
Local	256 GB pendrive	Weekly, and before any destructive step	<code>restic repo — data/raw/</code> , <code>db/</code> , <code>configs/</code> , git bundle
Off-site	Free-tier object store (Backblaze B2 gives 10 GB free; Cloudflare R2 gives 10 GB)	Nightly, incremental	Same restic repo, encrypted

```
restic -r /Volumes/BACKUP/institutional-research backup \
    data/raw db configs --exclude-caches
restic -r b2:institutional-research-backup backup data/raw db configs
restic check --read-data-subset=5%           # monthly: verify it can restore
```

restic rather than rsync because it is encrypted, deduplicated (2 GB of mostly-static parquet dedupes to near nothing on repeat runs), and — the part that matters — **verifiable**. A backup nobody has restored is a hypothesis. A quarterly restore drill into a scratch directory is a Phase 1 deliverable, not a later intention.

Free tiers are sufficient at 2 GB and require an owner account. Until that exists, the pendrive alone still removes the single-disk failure mode, so the cloud leg is not allowed to block Phase 1.

7. Definition of done for v1

The platform is complete when it can answer, with a re-derivable number and a recorded correction for how many times it looked:

1. Which institutions disclosed activity, and which are market makers rather than investors?
2. What happened after each disclosed transaction, at 9 horizons, against 6 benchmarks, net of a defensible cost model?
3. Does that differ between individual deals, accumulation, and consensus?
4. Does institutional *selling* carry different information from buying?
5. Are block deals different from bulk deals?
6. Does any participant show skill that survives Romano–Wolf correction — and does the same procedure on shuffled labels find just as many?
7. Does any seasonal pattern survive observation minimums, BY/BH/Storey, near-duplicate grouping, 1,000-rotation permutation, Hansen SPA, three-scheme out-of-sample, and full costs?
8. For every one of the above: which data version, which code commit, which inputs — reachable through the provenance DAG?

A "no" to questions 2 through 7, backed by 8, is a successful outcome.

8. Immediate next steps on approval

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|--|-----------------------|
| 1. Owner answers the remaining §5.2 questions | (~10 min) |
| 2. Owner plugs in the 256 GB pendrive | (Risk 8, local leg) |
| 3. Freeze MICCV2 – 3 launchd agents, tag | (~10 min, reversible) |
| 4. Repo scaffold + migration 0001 | (me) |
| 5. Copy v1_export, hash into the DAG | (me) |
| 6. restic backup + watched restore drill | (me) |
| 7. Rebuild the warehouse, run the §3.4 gate | (me) |
| 8. Report: files changed, schema, tests run, limitations, next steps | |

Nothing is deleted at step 2 — agents are unloaded and plists moved, both reversible. The first irreversible act in this plan is dropping `MICCV2/data/warehouse/`, and that happens only **after** the Phase 1 reconciliation gate proves the new warehouse reproduces it.